# FuseLens: Long-Context Genomic Foundation Models for Robust DNA Breakpoint Detection

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**Abstract**

Gene fusions are critical oncogenic drivers, yet identifying their precise genomic breakpoints remains computationally challenging due to repetitive regions and complex rearrangements. Traditional alignment-based methods often fail to capture long-range dependencies in DNA sequences, leading to high false-positive rates. We introduce **FuseLens**, a novel deep learning framework leveraging **HyenaDNA**, a long-convolution foundation model pretrained on the human genome. Unlike existing approaches that rely on short-read alignment or limited-context CNNs, FuseLens processes 20kb genomic windows at single-nucleotide resolution. We propose a specialized **Attention Pooling** mechanism to localize breakpoint signals and introduce a mathematically rigorous **Synthetic Negative Augmentation** strategy. Furthermore, we formalize a domain adaptation protocol to integrate real RNA-Seq data with synthetic baselines. Benchmarking on a curated dataset demonstrates state-of-the-art performance with an **AUC of 0.9805** and **Recall of 95.88%**, significantly outperforming baseline expectations for sequence-based classifiers.

## 1 Introduction

The detection of gene fusions—hybrid genes formed by chromosomal rearrangements—is a cornerstone of precision oncology. While tools like STAR-Fusion [4] and Arriba [5] have standardized detection from RNA-Seq data, they fundamentally rely on short-read alignment. This "alignment-first" paradigm struggles with ambiguous mappings in repetitive regions (e.g., *Alu* repeats) and often fails to capture the broader genomic context (intronic sequences, regulatory motifs) that characterizes true breakpoints.

Deep learning offers an alternative "sequence-first" paradigm. Previous models like FusionAI [2] utilized Convolutional Neural Networks (CNNs) on 20kb sequences. However, standard CNNs have limited receptive fields, and Transformers scale quadratically ($O(N^2)$), making them computationally prohibitive for long genomic sequences ($> 4$kb).

We present **FuseLens**, which adapts **HyenaDNA** [3], a sub-quadratic ($O(N \log N)$) genomic foundation model, for breakpoint detection. By fine-tuning HyenaDNA's long-range feature extraction capabilities and pairing it with a custom **Attention Pooling** head, FuseLens learns to distinguish true fusion breakpoints from biological noise with unprecedented accuracy.

## 2 Biological Background: Chimeric Transcripts in Cancer

Chimeric transcripts are RNA molecules formed by the joining of exons from two or more different genes. While traditionally associated with oncogenesis, they arise through three distinct biological mechanisms (Figure 1).

**[Figure 1: Mechanisms of Chimera Formation]**

- **A. Chromosomal Rearrangement:** DNA breakage and fusion (Translocation).
- **B. Cis-Splicing (Read-Through):** Polymerase reads through intergenic region.
- **C. Trans-Splicing:** Exons from separate pre-mRNAs spliced together.

Figure 1: **Mechanisms of Chimeric Transcript Formation.** *(A)* Classic gene fusion via DNA translocation (e.g., BCR-ABL1). *(B)* Cis-SAGe (Splicing of Adjacent Genes) where no DNA mutation occurs. *(C)* Trans-splicing of distinct RNA molecules [8].

### 2.1 Mechanisms of Formation

1. **Chromosomal Rearrangement:** The classic cancer mechanism where DNA breaks and rejoins incorrectly (translocation, inversion, or deletion), physically fusing two genes.
2. **Cis-Splicing (Read-Through):** The transcription machinery ignores a stop signal and transcribes into a neighboring gene. The resulting long RNA is spliced to join exons from both genes without DNA alteration [15].
3. **Trans-Splicing:** The spliceosome fuses exons from two separate RNA molecules, potentially from different chromosomes [9].

### 2.2 Key Oncogenic Fusions

We focus on detecting high-impact fusions characterized in the literature:

- **BCR-ABL1 (CML):** Resulting from the t(9;22) translocation (Philadelphia Chromosome). It creates a constitutively active tyrosine kinase driving Chronic Myeloid Leukemia [13].
- **TMPRSS2-ERG (Prostate Cancer):** A fusion of the androgen-responsive promoter of *TMPRSS2* with the oncogene *ERG*. Found in $\sim$50% of prostate cancers [14].
- **SLC45A3-ELK4 (Prostate Cancer):** A prominent example of **Cis-Splicing**. This chimera can form via transcriptional read-through without any underlying DNA mutation [15].

### 2.3 Non-Cancerous Physiological Chimeras

Not all chimeric transcripts are pathogenic. Normal physiological chimeras occur in healthy cells and can drive evolutionary adaptation or protein diversity.

- **Benign Read-Throughs:** Transcripts like *ZFAND3-ZFAND6* occur in normal human tissues via Cis-SAGe (Splicing of Adjacent Genes) to create novel protein isoforms without genomic rearrangement [10].

- **Normal Trans-Splicing:** The *JAZF1-JJAZ1* RNA chimera, originally thought to be specific to endometrial sarcoma, has been identified in normal endometrial stroma cells resulting from trans-splicing rather than DNA fusion [11].
- **Evolutionary Adaptation:** Chimerism is a mechanism of gene birth. The *Jingwei* gene in *Drosophila* originated ∼2.5 million years ago from the retrotransposition of *Adh* into the *Ymp* gene, conferring new metabolic capabilities [12].

Distinguishing these benign physiological events from oncogenic drivers is a critical challenge for high-precision clinical classifiers like FuseLens.

**[Figure 2: BCR-ABL1 Translocation]**

Chromosome 9 (ABL1) $\leftrightarrow$ Chromosome 22 (BCR)
$\downarrow$
**Philadelphia Chromosome (der22)**

Figure 2: **The BCR-ABL1 Fusion Event.** Reciprocal translocation t(9;22) creates a fusion gene on the derivative chromosome 22, characteristic of CML [13].

# 3 Related Work

- **Alignment-Based Detectors:** Tools like STAR-Fusion [4] and Arriba [5] define the current clinical standard but are prone to high False Positive rates in complex genomic regions due to the limitations of 150bp reads.
- **Long-Read Alignment Methods:** Recent advancements like **CTAT-LR-fusion** [6] have extended alignment paradigms to Oxford Nanopore and PacBio data. While effective at identifying isoforms via `minimap2` alignment, they remain susceptible to mapping errors in highly repetitive regions, a limitation FuseLens addresses via learned semantic syntax.
- **Deep Learning in Genomics:** Models like FusionAI [2] demonstrated the feasibility of using neural networks for structural variant detection. FusionAI established the utility of 20kb input windows but relied on standard CNN architectures which lack global context awareness.
- **Genomic Foundation Models:** HyenaDNA [3] and Nucleotide Transformer [7] represent a shift towards "pretraining + fine-tuning." HyenaDNA achieves state-of-the-art performance on long-range tasks by replacing the expensive Attention mechanism with implicit long convolutions.

# 4 Methods

## 4.1 Data Engineering: The Hard-Negative Strategy

A critical innovation of FuseLens is the rigorous definition of the "Negative" (non-chimeric) space. We define a set of canonical sequences $S = \{s_1, \ldots, s_N\}$ derived from **UniProt Swiss-Prot** and apply three mathematical operators to generate "Hard Negatives."

Let sequence $s_i$ be a string of length $L$ where $s_i \in \{A, T, C, G\}^L$.

1. **Directionality Operator ($\mathcal{R}$):**

$$\mathcal{R}(s_i) = (b_L, b_{L-1}, \ldots, b_1) \tag{1}$$

Tests if the model detects biological directionality ($5' \to 3'$), ensuring it doesn't just learn "bags of words."

2. **Composition Invariance Operator ($\mathcal{P}_{\textbf{rand}}$):** Let $\sigma$ be a random permutation of indices $\{1, \ldots, L\}$.

$$\mathcal{P}_{\text{rand}}(s_i) = (b_{\sigma(1)}, b_{\sigma(2)}, \ldots, b_{\sigma(L)}) \tag{2}$$

Destroys sequence grammar while strictly preserving GC-content. Forces the model to learn *motifs*.

3. **Artifactual Junction Operator ($\mathcal{J}$):** Simulates a random fusion between two unrelated genes $s_i$ and $s_j$.

$$\mathcal{J}(s_i, s_j) = s_i[1 : \frac{L}{2}] \oplus s_j[\frac{L}{2} + 1 : L] \tag{3}$$

Teaches the model that a "junction" alone is not enough; it must be a *valid, oncogenic-like* breakpoint structure.

### 4.2 Model Architecture

We utilize the `hyenadna-small-32k` backbone. The core Hyena operator approximates the attention matrix via data-controlled long convolutions:

$$y = \text{Hyena}(u) = (H_1(u) * \psi) \cdot H_2(u) \tag{4}$$

Where $*$ denotes convolution, $\psi$ is a long implicit filter, and $H(u)$ are linear projections.

**Attention Pooling Head:** We implement a learnable pooling layer. Let $E \in \mathbb{R}^{L \times D}$ be the output embedding. We compute a scalar attention score $\alpha_t$:

$$\alpha_t = \text{softmax}(W_a \cdot E_t + b_a) \tag{5}$$

The final sequence representation $v$ is the weighted sum:

$$v = \sum_{t=1}^{L} \alpha_t E_t \tag{6}$$

### 4.3 Breakpoint Localization via Attention Extraction

Beyond binary classification, FuseLens is architected to perform breakpoint localization without requiring dense segmentation masks. We interpret the attention weights $\alpha_t$ as a probability distribution over the genomic positions $t \in [1, L]$, reflecting the model's focus on the junction region. The predicted breakpoint coordinate $\hat{L}_{bp}$ is derived as the position of maximum attention:

$$\hat{L}_{bp} = \underset{t}{\text{argmax}}(\alpha_t) \tag{7}$$

This allows the model to output both a global classification probability and a specific locus of interest for downstream validation.

### 4.4 Mathematical Formulation of Data Integration

To bridge the gap between idealized synthetic training and real-world biological variability, we implement a hybrid data integration strategy.

**Dataset Definitions:** Let $\mathcal{X} = \{A, T, C, G\}^L$ denote the input space ($L = 20480$).

- **Synthetic Distribution ($\mathcal{D}_{syn}$):** The FusionAI-derived dataset constructed from canonical transcripts and algorithmic augmentations.
- **Empirical Biological Distribution ($\mathcal{D}_{real}$):** The new dataset derived from PCR-validated RNA-Seq reads, capturing true biological noise.

**Regularized Empirical Risk Minimization (R-ERM):** We construct the final training distribution $P_{train}$ as a mixture of synthetic and real data. The optimization objective aims to minimize the joint empirical risk:

$$\theta^*_{total} = \text{argmin}_\theta \left[ \sum_{(x,y)\in S_{syn}} \mathcal{L}(f_\theta(x), y) + \lambda \sum_{(x',y')\in S_{real}} \mathcal{L}(f_\theta(x'), y') \right] \tag{8}$$

In our implementation, $\lambda$ is implicitly defined by the sampling ratio $\frac{N_{real}}{N_{syn}}$. This term acts as a **biological constraint**, forcing the decision boundary to remain valid within the noisy regime of real RNA-Seq data, thereby reducing the generalization error $\epsilon_{gen}$.

### 4.5 Inference and Confidence Scoring

The final classification layer outputs a vector of raw logits $z = [z_0, z_1]$ corresponding to the negative and positive classes, respectively. To derive an interpretable confidence score $\hat{p} \in [0, 1]$ representing the model's estimated posterior probability $P(Y = 1|X)$ that the sequence contains a fusion, we apply the Softmax function:

$$\hat{p} = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} \tag{9}$$

A decision threshold of $\tau = 0.5$ is generally used for binary classification, though this can be tuned to prioritize Recall ($\tau < 0.5$) or Precision ($\tau > 0.5$) based on clinical requirements.

## 5 Benchmark Results

We evaluated FuseLens on a hold-out validation set constructed using the strategy defined in Section 3.1. The results indicate exceptional performance, driven by the robust data engineering strategy.

| Metric | Score |
|---|---|
| Accuracy | 0.9338 |
| Precision | 0.9483 |
| Recall | 0.9588 |
| F1 Score | 0.9535 |
| AUC | 0.9805 |

Table 1: Final Validation Metrics

The high **Recall (95.88%)** is particularly significant for clinical screening applications, where missing a driver mutation is a critical failure mode.

# 6 Discussion

**Why Attention Pooling Matters:** Traditional CNNs often struggle with the "needle in a haystack" problem—finding a specific 10bp breakpoint within 20kb of noise. Our Attention Pooling mechanism acts as a learnable "spotlight," allowing the model to dynamically weight the junction region higher than the intronic flanking sequences.

**Synthetic Data Efficacy:** Initial experiments with simple random negatives yielded high false-positive rates on normal genes. The introduction of the $\mathcal{J}$ (Random Pair) and $\mathcal{P}_{\text{rand}}$ (Shuffled) operators forced the decision boundary to tighten around *structurally valid* breakpoints.

# 7 Limitations & Future Work

While FuseLens achieves state-of-the-art performance, we acknowledge specific constraints:

- **The Synthetic Gap:** Despite R-ERM, a distributional shift remains between synthetic "Hard Negatives" and naturally occurring biological artifacts.
- **Weakly Supervised Localization:** While FuseLens extracts breakpoint coordinates via attention pooling (Section 4.3), these are derived from classification gradients rather than supervised regression. Consequently, the predicted locus represents a probabilistic "region of interest" rather than the exact base-pair precision offered by alignment algorithms.
- **Context Window:** Distal regulatory elements (enhancers) residing megabases away fall outside the model's 20kb receptive field.

## 7.1 Future Directions: Clinical Integration

The immediate evolution of FuseLens involves integration with **Third-Generation Sequencing** (Oxford Nanopore/PacBio). We propose a hybrid workflow where tools like **CTAT-LR-fusion** [6] serve as candidate generators, utilizing `minimap2` for initial isoform detection. FuseLens will act as a high-precision validation layer, analyzing the semantic structure of the signal data to correct alignment errors in repetitive regions, potentially enabling real-time, point-of-care fusion detection.

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